

# Olympiad Test Paper — Science (Class 7) — Paper 4

## Chapter 13: Motion and Time

### Paper 4 — (progressively harder)

|                        |                                      |
|------------------------|--------------------------------------|
| <b>Subject</b>         | Science (NCERT Class 7)              |
| <b>Chapter</b>         | 13 — Motion and Time                 |
| <b>Total Questions</b> | 60 (Multiple Choice, single correct) |
| <b>Maximum Marks</b>   | 240                                  |
| <b>Suggested Time</b>  | 90 minutes                           |

**Marking scheme:** +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

**Instructions:** Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors weaponise the plain-mean average-speed error, the wrong-direction unit conversion, slope-vs-y-value graph confusion, harmonic-mean traps over equal distances, pendulum period blamed on mass/amplitude, rest segments counted as travel, and cumulative-graph 'return' read as extra distance. Do not write on this paper — mark answers on your answer sheet.

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**1.** A spinning top stays in one place while every point on it traces circles about the central spike, and the top returns to the same orientation at regular intervals. The two descriptors that best fit a point on its rim are:

- (A) oscillatory and periodic                      (B) rectilinear and periodic  
(C) circular and oscillatory                      (D) circular and periodic

**2.** A robot covers 144 m in the first 12 s and an additional 96 m in the next 8 s, keeping a steady pace throughout. Its single steady speed is:

- (A) 10 m/s                      (B) 11 m/s  
(C) 20 m/s                      (D) 12 m/s

**3.** A peregrine falcon dives at 360 km/h. Expressed in metres per second this is:

- (A) 1296 m/s                      (B) 10 m/s  
(C) 100 m/s                      (D) 60 m/s

**4.** A hiker walks the first one-third of a trail at 2 km/h, the middle one-third at 3 km/h and the last one-third at 6 km/h, with the three parts being equal in length. The average speed for the whole trail is:

- (A) 3 km/h  
(C) 4 km/h
- (B) 3.67 km/h  
(D) 3.5 km/h

**5.** A delivery van drives a fixed route out at 50 km/h and back along the same road at 75 km/h. Its average speed over the whole there-and-back trip is:

- (A) 62.5 km/h  
(C) 60 km/h
- (B) 65 km/h  
(D) 125 km/h

**6.** Two pendulums hang from the same beam. Pendulum X has a 1 m string and a heavy brass bob; pendulum Y has a 1 m string and a light cork bob, and Y is released from a much wider angle. Compared with X, the time period of Y is:

- (A) longer, because Y is released from a wider angle  
(B) shorter, because the cork bob is lighter  
(C) longer, because the brass bob is heavier so X swings slower  
(D) the same, because period depends on the string length, not the bob mass or the release angle

**7.** A driver's dashboard shows '60' on a dial that constantly rises and falls with the accelerator, and a separate set of digits that only ever increases and had advanced by 320 since morning. These two instruments are, in order:

- (A) the odometer (current speed) and the speedometer (total distance)  
(B) two speedometers reading the same speed twice  
(C) the speedometer (showing current speed) and the odometer (showing total distance)  
(D) a stopwatch and the speedometer

**8.** A swift flies at 30 m/s. Converting to km/h and then identifying the faster of this and a train at 100 km/h:

- (A) the train is faster; the swift is only 8.3 km/h  
(B) they are equal once converted  
(C) the swift is faster; it is 108 km/h versus 100 km/h  
(D) the swift is faster at 180 km/h

**9.** A beetle crawls 24 cm in 8 s, then pauses for 4 s, then crawls a further 12 cm in 6 s along a straight line. Its average speed over the whole observation, in m/s, is:

- (A) 0.02 m/s  
(C) 0.04 m/s
- (B) 0.03 m/s  
(D) 0.06 m/s

**10.** On a distance-time graph an object's line slopes up from (0 s, 0 m) to (4 s, 40 m), runs horizontally from (4 s, 40 m) to (10 s, 40 m), then slopes up again from (10 s, 40 m) to (12 s, 70 m). During the steepest part its speed is:

- (A) 10 m/s, during the first segment from 0 s to 4 s  
(B) 40 m/s, read off the flat middle segment  
(C) 5 m/s, averaged across the whole 12 s

(D) 15 m/s, during the last segment from 10 s to 12 s

**11.** A train covers 40 km in the 1st hour, 40 km in the 2nd hour and 40 km in the 3rd hour along a straight track. Choose the most precise true statement:

- (A) the motion is non-uniform, because the total distance keeps growing
- (B) the motion is uniform, because equal distances are covered in equal time intervals
- (C) the motion is circular, because trains follow curved rails
- (D) the speed is increasing, because more total distance has built up by hour 3

**12.** A sewing-machine needle moves straight up and down through the cloth, repeating its stroke at a fixed rate. The single best description of the needle's motion is:

- (A) oscillatory (and therefore periodic)
- (B) circular
- (C) rectilinear and non-periodic
- (D) random

**13.** On a distance-time graph, line P runs from (0,0) to (5 s, 45 m) and line Q runs from (0,0) to (5 s, 30 m), both straight. The speed of P exceeds the speed of Q by:

- (A) 3 m/s
- (B) 15 m/s
- (C) 1.5 m/s
- (D) 9 m/s

**14.** A motorist wants a 360 km trip to average 80 km/h. The first 240 km were driven at 60 km/h. What constant speed over the remaining 120 km would just achieve the planned average?

- (A) 100 km/h
- (B) 120 km/h
- (C) 240 km/h
- (D) it is impossible at any finite speed

**15.** A pendulum completes 45 oscillations in 90 s. Its string is then shortened so that its time period becomes exactly half of the original. How many oscillations does it now complete in 90 s?

- (A) 45
- (B) 22
- (C) 90
- (D) 180

**16.** Among the following four speeds, the single greatest is:

- (A) 20 m/s
- (B) 84 km/h
- (C) 0.4 km/min
- (D) 1500 m/min

**17.** The SI unit of time is the second. Which ordering correctly lists these durations from shortest to longest?

- (A) 1 hour < 1 minute < 1 day < 1 week
- (B) 1 minute < 1 hour < 1 week < 1 day
- (C) 1 minute < 1 hour < 1 day < 1 week
- (D) 1 day < 1 week < 1 hour < 1 minute

**18.** On a distance-time graph the line is straight from (0 s, 0 m) to (8 s, 80 m), horizontal from (8 s, 80 m) to (14 s, 80 m), then straight back down from (14 s, 80 m) to (20 s, 0 m). The total path length travelled over the 20 s is:

- (A) 80 m
- (B) 0 m
- (C) 240 m
- (D) 160 m

**19.** An object's distance-time graph passes through (3 s, 12 m) and (9 s, 42 m) along a straight line. Reading its speed:

- (A) 7 m/s
- (B) 5 m/s
- (C) 4 m/s
- (D) 6 m/s

**20.** A bus runs 50 km in the 1st hour, 60 km in the 2nd hour and 70 km in the 3rd hour, all on a straight road. Its average speed over the three hours, and the nature of its motion, are:

- (A) 60 km/h, and the motion is uniform
- (B) 180 km/h, and the motion is uniform
- (C) 60 km/h, and the motion is non-uniform
- (D) 50 km/h, and the motion is non-uniform

**21.** A boat goes 36 km downstream in 2 h and the same 36 km upstream in 3 h. Treating the whole there-and-back run, its average speed is:

- (A) 15 km/h
- (B) 14.4 km/h
- (C) 18 km/h
- (D) 12 km/h

**22.** A wall clock's minute hand sweeps once around the dial every hour, returning to the 12 each time. The two descriptors that fit the tip of the minute hand are:

- (A) oscillatory and periodic
- (B) circular and periodic
- (C) rectilinear and periodic
- (D) random and circular

**23.** A pendulum on Earth has a time period of exactly 2 s. To shorten its period to 1 s, its length should be changed to:

- (A) one-half of its original length
- (B) double its original length
- (C) one-quarter of its original length
- (D) four times its original length

**24.** Spot the INCORRECT speed conversion among these four:

- (A) 108 km/h = 36 m/s
- (B) 18 km/h = 5 m/s
- (C) 45 km/h = 12.5 m/s
- (D) 144 km/h = 40 m/s

**25.** On a distance-time graph two straight lines start at different heights and cross at the point (6 s, 30 m). At the instant of crossing the two objects:

- (A) are moving at the same speed
- (B) both momentarily stop
- (C) swap their directions of travel
- (D) are at the same distance from the start at that moment

**26.** A metronome arm ticks 30 times in 20 s at a steady beat. Using these ticks to mark equal time intervals is reliable because the motion is:

- (A) periodic, so each interval between ticks is the same
- (B) non-uniform, which spreads the ticks out usefully
- (C) random, so averaging many ticks helps
- (D) rectilinear, so the ticks travel in a straight line

**27.** A car covers 90 km, then is parked (at rest) for 30 minutes, then covers a further 90 km. The first 90 km took 1.5 h and the second 90 km took 1 h. The average speed for the entire outing, including the parked time, is:

- (A) 60 km/h
- (B) 72 km/h
- (C) 90 km/h
- (D) 45 km/h

**28.** A sand clock (hourglass) is trusted to measure a fixed interval because:

- (A) the sand trickles through the narrow neck at a roughly steady rate
- (B) the sand grains push each other in regular pulses
- (C) the two glass bulbs swing to-and-fro like a pendulum
- (D) sunlight warms the upper bulb on a fixed schedule

**29.** A motorbike travels at a steady 72 km/h. How many metres does it cover in 90 seconds?

- (A) 6480 m
- (B) 1800 m
- (C) 1080 m
- (D) 108 m

**30.** On a distance-time graph an object's line gets less and less steep while always continuing to rise. Over this stretch the object is:

- (A) moving backward toward the start
- (B) still moving forward, but slowing down
- (C) speeding up the whole time
- (D) at rest after the bend

**31.** A pendulum makes 24 oscillations in 36 s. The frequency, and the way to read it, are:

- (A)  $\frac{2}{3}$  oscillation per second, found as oscillations divided by total time
- (B) 1.5 oscillations per second, found as time divided by oscillations
- (C) 24 oscillations per second, the count itself
- (D) 36 oscillations per second, the time itself

**32.** To find a vehicle's true average speed for a complete journey, the most direct instruments are:

- (A) the speedometer read once at the journey's end
- (B) the odometer for total distance and a clock for total time
- (C) the speedometer read once at the journey's start
- (D) a sundial alone

**33.** A toy car rolls 7 m in the 1st second, 5 m in the 2nd second and 3 m in the 3rd second along a straight floor. Its motion is:

- (A) uniform, because it keeps moving forward
- (B) non-uniform, slowing down
- (C) non-uniform, speeding up
- (D) at rest, because the distances are small

**34.** Light travels about 300000 km in 1 s. Sunlight takes roughly 500 s to reach Earth. The Sun is therefore about:

- (A) 600 million km away
- (B) 15 million km away
- (C) 150 million km away
- (D) 300000 km away

**35.** Two cars start from the same point. On a distance-time graph car A's line is a straight slant from the origin and car B's line bends upward from the origin, starting gentler than A. Which statement can be correct?

- (A) A is always ahead because a straight line always beats a curve
- (B) B is always ahead because a curve always wins
- (C) B starts slower than A but can overtake and finish ahead if its line bends steeply enough
- (D) they stay at the same point throughout since both start at the origin

**36.** A runner covers 200 m in 25 s, then a further 300 m in 75 s. The average speed over the full 500 m is:

- (A) 6 m/s
- (B) 4 m/s
- (C) 10 m/s
- (D) 5 m/s

**37.** On a distance-time graph, line A goes from (0,0) to (8 s, 120 m) and line B goes from (0,0) to (8 s, 72 m). The ratio of A's speed to B's speed is:

- (A) 3 : 5
- (B) 120 : 72 and cannot be simplified
- (C) 1 : 1 because the times match
- (D) 5 : 3

**38.** On a distance-time graph the line is perfectly horizontal at a height of 50 m from  $t = 4$  s to  $t = 11$  s. During this stretch the object's speed is:

- (A) zero, because its distance does not change with time
- (B) 50 m/s, read straight off the height
- (C) very high, since a flat line means smooth fast travel
- (D) 7 m/s, the width of the segment

**39.** Two pendulums hang side by side: one is 1 m long, the other 9 m long. Over one minute, comparing the number of oscillations:

- (A) the 9 m pendulum makes more, because a longer string swings more often
- (B) the 1 m pendulum makes more oscillations, because the shorter pendulum has the smaller period
- (C) they make the same number, since mass is not stated
- (D) the 9 m pendulum makes exactly 9 times as many

**40.** A flag on a pole flutters back and forth in the wind without settling into any fixed rhythm. Its motion is best described as:

- (A) to-and-fro but non-periodic, since it does not repeat at fixed intervals
- (B) oscillatory and periodic, like a clock pendulum
- (C) circular, because the flag wraps around
- (D) rectilinear, because the pole is straight

**41.** A car covers 100 km, idles in traffic for 20 minutes, then covers a further 140 km. The moving stretches took 1 h and 2 h respectively. The average speed for the whole trip, including the idle, is:

- (A) 80 km/h  
(C) 60 km/h
- (B) 72 km/h  
(D) 120 km/h

**42.** A cyclist walks 6 km/h going to a park 3 km away and pedals back the same 3 km at 18 km/h. The average speed for the round trip is:

- (A) 9 km/h  
(C) 10 km/h
- (B) 12 km/h  
(D) 9.5 km/h

**43.** An ant covers 36 mm in 6 s. Expressed in m/s, its speed is:

- (A) 6 m/s  
(C) 0.06 m/s
- (B) 0.006 m/s  
(D) 0.6 m/s

**44.** A pendulum makes 36 complete oscillations in 54 s. How many complete oscillations does it make in 3 minutes at the same rate?

- (A) 108  
(C) 162
- (B) 72  
(D) 120

**45.** On a distance-time graph an object's line rises from (0 s, 0 m) to (3 s, 30 m), is flat from (3 s, 30 m) to (8 s, 30 m), then rises from (8 s, 30 m) to (11 s, 90 m). The fastest segment, and its speed, are:

- (A) the first segment, at 10 m/s  
(C) the last segment, at 20 m/s
- (B) the flat middle segment, at 30 m/s  
(D) they are all equal at 30 m/s

**46.** A water clock keeps time by the steady drip of water from one vessel into another. Which of these would make it UNRELIABLE compared with a sundial?

- (A) a cloudy sky overhead  
(B) nothing in this list; unlike a sundial, a water clock works at night and on cloudy days  
(C) the arrival of nighttime  
(D) the Sun being low on the horizon

**47.** Three messengers cover the same 12 km. P takes 30 minutes, Q takes 20 minutes and R takes 40 minutes. Ranking their speeds from fastest to slowest gives:

- (A) P, then Q, then R  
(C) R, then P, then Q
- (B) Q, then P, then R  
(D) they are all equal because the distance is the same

**48.** A car drives the first 60 km at 30 km/h, the next 60 km at 60 km/h and the last 60 km at 90 km/h. The average speed for the whole 180 km is closest to:

- (A) 60 km/h  
(C) 45 km/h
- (B) 55 km/h  
(D) 49 km/h

**49.** A high-speed train runs at 252 km/h. How far does it travel in 25 seconds?

- (A) 6300 m  
(C) 1750 m
- (B) 175 m  
(D) 1050 m

**50.** Which statement about distance-time graphs is FALSE?

- (A) a steeper straight line means a faster object
- (B) the slope of the line gives the speed
- (C) a horizontal line means the object is moving at a steady high speed
- (D) a straight slanted line means uniform speed

**51.** A swimmer covers 1.8 km in 30 minutes at a steady pace. Her average speed in m/s is:

- (A) 3.6 m/s
- (B) 0.06 m/s
- (C) 60 m/s
- (D) 1 m/s

**52.** A pendulum clock that keeps perfect time has its bob slid down so the string is effectively made longer. From then on, compared with correct time, the clock will:

- (A) run fast, because a longer pendulum swings more often
- (B) keep correct time, since period depends only on the bob's mass
- (C) run slow, because a longer pendulum has a longer period and so ticks less often
- (D) stop, because a long pendulum cannot swing

**53.** On a distance-time graph (distance measured from the start) an object's line climbs to a peak at (10 s, 60 m) and then falls back to the time axis at (16 s, 0 m). This means the object:

- (A) moved away to 60 m and then returned to its starting point
- (B) kept speeding up the entire time
- (C) moved in a circle and disappeared
- (D) stopped permanently at the 60 m peak

**54.** On a distance-time graph object M's straight line passes through (1 s, 10 m) and (5 s, 30 m). Its speed is:

- (A) 6 m/s
- (B) 5 m/s
- (C) 10 m/s
- (D) 30 m/s

**55.** A walker sets out at 9:00 toward a town 24 km away at a steady 6 km/h. A cyclist leaves the same point along the same road at 11:00 and rides at 12 km/h. The cyclist catches up with the walker at:

- (A) 1:00, after the cyclist has ridden 24 km
- (B) 12:00, after the cyclist has ridden 12 km
- (C) 11:30, after the cyclist has ridden 6 km
- (D) they never meet before the town

**56.** A train 80 m long passes a stationary signal post completely in 4 s. Its speed is:

- (A) 4 m/s
- (B) 20 m/s
- (C) 320 m/s
- (D) 5 m/s

**57.** Two cars start from the same town. On a distance-time graph car A's straight line from the origin is steeper than car B's straight line from the origin. After any time has elapsed:

- (A) car A is always farther from the town than car B
- (B) car B is always farther because its line is gentler
- (C) they stay exactly the same distance apart at all times
- (D) car A is behind car B until their lines cross later

**58.** Spot the INCORRECT speed conversion among these:

(A)  $36 \text{ km/h} = 10 \text{ m/s}$

(B)  $90 \text{ km/h} = 25 \text{ m/s}$

(C)  $126 \text{ km/h} = 35 \text{ m/s}$

(D)  $63 \text{ km/h} = 18.5 \text{ m/s}$

**59.** An odometer reads 24,300 km at the start of a drive and 24,540 km at the end; the drive lasted 3 hours including a 30 minute break. The car's average speed for the whole drive was:

(A)  $96 \text{ km/h}$

(B)  $60 \text{ km/h}$

(C)  $78 \text{ km/h}$

(D)  $80 \text{ km/h}$

**60.** On a distance-time graph an object's straight line passes through (2 s, 14 m) and (9 s, 49 m). Its speed is:

(A)  $7 \text{ m/s}$

(B)  $5.4 \text{ m/s}$

(C)  $6 \text{ m/s}$

(D)  $5 \text{ m/s}$

# Solutions — Science (Class 7), Chapter 13:

## Motion and Time — Paper 4

### Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

### Answer Key

| Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans | Q  | Ans |
|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|
| 1  | D   | 11 | B   | 21 | B   | 31 | A   | 41 | B   | 51 | D   |
| 2  | D   | 12 | A   | 22 | B   | 32 | B   | 42 | A   | 52 | C   |
| 3  | C   | 13 | A   | 23 | C   | 33 | B   | 43 | B   | 53 | A   |
| 4  | A   | 14 | C   | 24 | A   | 34 | C   | 44 | D   | 54 | B   |
| 5  | C   | 15 | C   | 25 | D   | 35 | C   | 45 | C   | 55 | A   |
| 6  | D   | 16 | D   | 26 | A   | 36 | D   | 46 | B   | 56 | B   |
| 7  | C   | 17 | C   | 27 | A   | 37 | D   | 47 | B   | 57 | A   |
| 8  | C   | 18 | D   | 28 | A   | 38 | A   | 48 | D   | 58 | D   |
| 9  | A   | 19 | B   | 29 | B   | 39 | B   | 49 | C   | 59 | D   |
| 10 | D   | 20 | C   | 30 | B   | 40 | A   | 50 | C   | 60 | D   |

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

### Worked Solutions

**1. (D)** A rim point sweeps a closed circle, so the motion is circular, and because it repeats the same path at fixed intervals it is periodic. It never reverses to-and-fro about a mean position, so it is not oscillatory; nor does it travel along a straight line, so rectilinear is wrong.

**2. (D)** Total path is  $144 + 96 = 240$  m in  $12 + 8 = 20$  s, giving  $240/20 = 12$  m/s; the same value drops out of either leg alone ( $144/12$  and  $96/8$ ), confirming the pace is steady. Averaging the two leg speeds or dividing only one leg by the whole time gives the wrong numbers.

**3. (C)** Multiply by  $5/18$ :  $360 \times 5/18 = 100$  m/s. Multiplying by 3.6 instead (1296) reverses the conversion, and dividing by 36 (10) or by 6 (60) applies a wrong factor.

**4. (A)** Take 6 km of trail so each leg is 2 km. Times are  $2/2 = 1$  h,  $2/3$  h and  $2/6 = 1/3$  h, totalling  $1 + 2/3 + 1/3 = 2$  h for 6 km, so  $6/2 = 3$  km/h. The plain mean of 2, 3 and 6 (3.67) is the equal-distance trap.

- 5. (C)** Let each way be 150 km: out takes 3 h, back takes 2 h, so 300 km in 5 h gives 60 km/h. The plain mean (62.5) ignores that more time is spent on the slower leg.
- 6. (D)** For a simple pendulum the period is set by the length alone; equal 1 m strings give equal periods. The bob's mass and the (small) release angle do not change it, so the mass and amplitude distractors are wrong.
- 7. (C)** The dial that swings up and down tracks the instantaneous speed, so it is the speedometer; the counter that only climbs records cumulative distance, so it is the odometer. The swapped-label option reverses the two instruments.
- 8. (C)**  $30 \text{ m/s} \times 3.6 = 108 \text{ km/h}$ , which exceeds the train's 100 km/h, so the swift wins. Dividing by 3.6 (8.3) or multiplying by 6 (180) misconverts the speed.
- 9. (A)** Total distance  $24 + 12 = 36 \text{ cm} = 0.36 \text{ m}$  over total time  $8 + 4 + 6 = 18 \text{ s}$  gives  $0.36/18 = 0.02 \text{ m/s}$ . Forgetting the 4 s pause ( $0.36/14 \sim 0.026$ , rounded toward 0.03) or ignoring one leg inflates the figure.
- 10. (D)** The last segment rises  $70 - 40 = 30 \text{ m}$  in  $12 - 10 = 2 \text{ s}$ , a slope of 15 m/s, which is steeper than the first segment's  $40/4 = 10 \text{ m/s}$ . The flat part is at rest, and reading 40 off the axis instead of using the slope is the y-value trap.
- 11. (B)** Equal 40 km chunks in equal 1 h intervals mean the speed is constant, the definition of uniform motion along a straight track. A growing cumulative total or a longer total path does not make speed change, and the rails here are straight.
- 12. (A)** The needle repeats a to-and-fro stroke about a mean position at a fixed rate, which is oscillatory and hence also periodic. It does not trace a circle, and because the stroke repeats it is not non-periodic or random.
- 13. (A)** P's slope is  $45/5 = 9 \text{ m/s}$  and Q's is  $30/5 = 6 \text{ m/s}$ , a difference of 3 m/s. Subtracting the distances ( $45 - 30 = 15$ ) ignores that you must compare slopes, not endpoint heights.
- 14. (C)** The whole trip must take  $360/80 = 4.5 \text{ h}$ . The first 240 km at 60 km/h already used  $240/60 = 4 \text{ h}$ , leaving only 0.5 h for the last 120 km, so  $120/0.5 = 240 \text{ km/h}$ . Splitting the remaining distance by the target speed (100 or 120) ignores how little time is left.
- 15. (C)** Originally  $90/45 = 2 \text{ s}$  per oscillation; halving the period gives 1 s each, so in 90 s it makes  $90/1 = 90$  oscillations. Leaving the count unchanged ignores the shorter period, and 180 doubles a step too far.
- 16. (D)** Convert every option to m/s.  $1500 \text{ m/min} = 1500/60 = 25 \text{ m/s}$ .  $84 \text{ km/h} = 84000/3600 \sim 23.3 \text{ m/s}$ , just below. 20 m/s is given directly.  $0.4 \text{ km/min} = 400 \text{ m/min} \sim$

6.7 m/s. The largest value is 25 m/s, so the fastest speed is 1500 m/min; the 84 km/h figure is a deliberately close runner-up that still converts to under 24 m/s, and the bare 20 m/s and the slow 0.4 km/min are clearly smaller.

**17. (C)**  $60 \text{ s} < 3600 \text{ s} < 86400 \text{ s} < 604800 \text{ s}$ , so minute, hour, day, week is the increasing order. The other lists swap an hour below a minute or a week below a day.

**18. (D)** The object goes out 80 m, waits, then returns 80 m, so the path covered is  $80 + 80 = 160 \text{ m}$  even though it ends at the start. Reading 80 m counts only the outward trip, and 0 m mistakes net displacement for distance travelled.

**19. (B)** Speed is the slope,  $(42 - 12)/(9 - 3) = 30/6 = 5 \text{ m/s}$ . Dividing 42 by 6 (7) uses the wrong rise, and  $42/9$  mixes a y-value with the wrong time.

**20. (C)** Total  $50 + 60 + 70 = 180 \text{ km}$  in 3 h gives an average of 60 km/h, but since unequal distances were covered in equal time intervals the motion is non-uniform. Calling it uniform contradicts the changing hourly distances; 180 is the total, not the average.

**21. (B)** Total distance  $36 + 36 = 72 \text{ km}$  over total time  $2 + 3 = 5 \text{ h}$  gives  $72/5 = 14.4 \text{ km/h}$ . The plain mean of the two leg speeds (18 and 12 give 15) over-counts the faster leg's contribution.

**22. (B)** The tip traces a complete circle and returns to the same spot at fixed intervals, making it circular and periodic. It never reverses to-and-fro (so not oscillatory), does not move in a straight line, and is far from random.

**23. (C)** Period scales with the square root of length, so to halve the period the length must be quartered (because the square root of  $1/4$  is  $1/2$ ). Halving the length only shortens the period by the square root of  $1/2$ , not to half.

**24. (A)**  $108 \times 5/18 = 30 \text{ m/s}$ , not 36, so that line is wrong. The other three check out:  $18 \times 5/18 = 5$ ,  $45 \times 5/18 = 12.5$ , and  $144 \times 5/18 = 40$ .

**25. (D)** A crossing point shares the same time and the same distance value, so both objects are 30 m from the start at 6 s. Equal positions say nothing about their slopes, so their speeds need not match, and neither stops nor reverses.

**26. (A)** A periodic motion repeats at fixed intervals, so equal gaps between ticks make a dependable timekeeper. Random or non-uniform ticking would give unequal intervals, and the rectilinear label describes a path, not timing.

**27. (A)** Total distance  $90 + 90 = 180 \text{ km}$  over total time  $1.5 + 0.5 + 1 = 3 \text{ h}$  gives 60 km/h. Leaving out the 0.5 h park ( $180/2.5 = 72$ ) is the trap of not counting rest time.

**28. (A)** The constant trickle of sand through the neck takes the same time each run, which is what makes the interval repeatable. The bulbs do not swing, the grains do not pulse, and

sunlight plays no part.

**29. (B)**  $72 \text{ km/h} = 20 \text{ m/s}$ , and  $20 \times 90 = 1800 \text{ m}$ . Multiplying 72 by 90 directly (6480) forgets the unit conversion, while 1080 and 108 use the wrong factor.

**30. (B)** A rising line means distance keeps increasing, so the object moves forward; a slope that flattens means the speed is dropping. A falling line would be needed for backward motion, and a horizontal line for rest.

**31. (A)** Frequency is the number of oscillations per second,  $24/36 = 2/3$  per second. Dividing time by oscillations ( $36/24 = 1.5$ ) gives the period, not the frequency, and the raw counts are not rates.

**32. (B)** Average speed is total distance over total time, so you read distance from the odometer and time from a clock. A single speedometer glance gives only one instant's speed, and a sundial measures neither distance nor a precise short interval.

**33. (B)** Each successive second covers less ground (7, then 5, then 3 m), so the speed is falling: non-uniform and slowing. Moving forward does not make it uniform, and shrinking gaps rule out speeding up.

**34. (C)** Distance is speed times time,  $300000 \times 500 = 150000000 \text{ km}$ , i.e. 150 million km. Using 300000 alone ignores the travel time, and the other figures misplace the factor of 500.

**35. (C)** A's constant slope competes with B's steepening slope; since B's line bends upward it can become steeper than A and cross above it, ending ahead. Neither shape guarantees the lead the whole way, and equal starts do not mean equal positions thereafter.

**36. (D)** Total  $200 + 300 = 500 \text{ m}$  over  $25 + 75 = 100 \text{ s}$  gives  $500/100 = 5 \text{ m/s}$ . Averaging the two leg speeds (8 m/s and 4 m/s gives 6) over-weights the short fast leg.

**37. (D)** Slopes are  $120/8 = 15 \text{ m/s}$  and  $72/8 = 9 \text{ m/s}$ , whose ratio 15 : 9 simplifies to 5 : 3. The inverted ratio swaps the objects, and equal times do not make equal speeds.

**38. (A)** A flat line means the distance stays at 50 m, so no ground is covered and the speed is zero. Reading the 50 m height as a speed is the y-value trap, and the 7 s width is a time, not a speed.

**39. (B)** A shorter pendulum has a shorter period, so it completes more swings per minute; the 1 m one wins. A longer string actually swings less often, and the count does not scale directly with length.

**40. (A)** The flag does move back and forth, but the gusts make the intervals irregular, so it is not periodic. A clock pendulum, by contrast, repeats at fixed intervals; the flag neither traces a circle nor moves in a straight line.

- 41. (B)** Total distance  $100 + 140 = 240$  km over total time  $1 + 1/3 + 2 = 10/3$  h gives  $240 / (10/3) = 72$  km/h. Ignoring the 20 min idle ( $240/3 = 80$ ) over-states the average.
- 42. (A)** Out takes  $3/6 = 0.5$  h, back takes  $3/18 = 1/6$  h, so total 6 km in  $0.5 + 1/6 = 2/3$  h gives  $6 / (2/3) = 9$  km/h. The plain mean of 6 and 18 (12) is the equal-distance trap.
- 43. (B)** 36 mm = 0.036 m, and  $0.036/6 = 0.006$  m/s. Forgetting that millimetres are thousandths of a metre shifts the decimal point and yields the larger wrong values.
- 44. (D)** Its period is  $54/36 = 1.5$  s, so in 3 minutes = 180 s it makes  $180/1.5 = 120$  oscillations. Reusing the 36 count or scaling by the wrong factor gives the other figures.
- 45. (C)** The last segment climbs  $90 - 30 = 60$  m in  $11 - 8 = 3$  s, a slope of 20 m/s, beating the first segment's  $30/3 = 10$  m/s; the flat part is at rest. Reading the 30 m height as a speed is the y-value trap.
- 46. (B)** A water clock depends only on the steady flow of water, so darkness and clouds do not stop it; those conditions defeat a sundial, which needs sunlight. Hence none of the sky-and-Sun conditions disable a water clock.
- 47. (B)** For a fixed distance, less time means more speed, so the order of increasing time (Q 20 min, P 30 min, R 40 min) reverses to the order of decreasing speed: Q fastest, then P, then R. Equal distance does not mean equal speed when times differ.
- 48. (D)** Times are  $60/30 = 2$  h,  $60/60 = 1$  h and  $60/90 = 2/3$  h, totalling  $11/3$  h for 180 km, so  $180 / (11/3) =$  about 49 km/h. The plain mean of 30, 60 and 90 (60) is the equal-distance trap.
- 49. (C)**  $252 \text{ km/h} = 252 \times 5/18 = 70$  m/s, and  $70 \times 25 = 1750$  m. Multiplying 252 by 25 directly (6300) skips the conversion, and the other figures use the wrong factor.
- 50. (C)** A horizontal line means the distance is not changing, so the object is at rest, not moving fast; that statement is false. The other three correctly describe slope as speed, steeper as faster, and a straight slant as uniform motion.
- 51. (D)**  $1.8 \text{ km} = 1800 \text{ m}$  and  $30 \text{ min} = 1800 \text{ s}$ , so  $1800/1800 = 1$  m/s. Treating 30 min as 30 s (60 m/s) or as 0.5 h without converting to m/s yields the wrong values.
- 52. (C)** Lengthening the pendulum lengthens its period, so each tick takes longer and the clock falls behind. Mass does not set the period, and a longer pendulum certainly still swings, just more slowly.
- 53. (A)** Returning to distance zero on a from-start graph means the object came back to where it began. The rise-then-fall shape shows out-and-back motion, not continuous speeding up or a permanent stop, and circles are not represented this way.

**54. (B)** Slope is  $(30 - 10)/(5 - 1) = 20/4 = 5$  m/s. Dividing 30 by 5 (6) or reading the final y-value (30) ignores that you need the change in distance over the change in time.

**55. (A)** By 11:00 the walker has covered  $6 \times 2 = 12$  km and keeps gaining 6 km each hour, while the cyclist gains 12 km each hour, closing the 12 km gap at  $12 - 6 = 6$  km/h, which takes  $12/6 = 2$  h, i.e. at 1:00; the cyclist has then ridden  $12 \times 2 = 24$  km, exactly reaching the town. The earlier times under-count the gap or the closing rate.

**56. (B)** To clear a post, the train travels its own length, 80 m, in 4 s, giving  $80/4 = 20$  m/s. Multiplying length by time (320) or dividing the wrong way gives the other figures.

**57. (A)** Both lines start together at the origin, and the steeper one rises faster, so car A is always ahead and the gap widens; the lines never cross again. A gentler line cannot be farther, and the gap is not constant.

**58. (D)**  $63 \times 5/18 = 17.5$  m/s, not 18.5, so that one is wrong. The others are correct: 36, 90 and 126 times  $5/18$  give exactly 10, 25 and 35 m/s.

**59. (D)** Distance is  $24540 - 24300 = 240$  km over the full 3 h, giving  $240/3 = 80$  km/h. Subtracting the break from the time ( $240/2.5 = 96$ ) ignores that average speed uses the total elapsed time, break included.

**60. (D)** Slope is  $(49 - 14)/(9 - 2) = 35/7 = 5$  m/s. Dividing 49 by 7 (7) uses the wrong rise, and reading  $49/9$  mixes a y-value with the wrong time.